Between and Within: Income, Labour Markets, and Inter-State Migration in India

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Abstract

This paper examines the correlates of inter-state migration in India using state-level panel data from the Census of India and the Reserve Bank of India, covering three Census years -1991, 2001, and 2011. Migration is measured as the rate of inter-state migrants relative to state population. The analysis employs two complementary estimators: a between-state estimator that identifies associations from persistent cross-state differences, and a two-way fixed effects estimator that exploits within-state variation over time.

Between states, per capita income is the dominant predictor of migration, explaining approximately 53 percent of cross-state variation. The income coefficient remains stable and significant across sequential specifications, confirming that it operates independently rather than as a proxy for urbanization or density. Within states over time, labour force participation and urbanization are positively and statistically significantly associated with migration, while social expenditure share and population density do not exhibit robust independent associations. The within-group R^2 is close to zero, indicating that persistent cross-state differences absorbed by state fixed effects account for the dominant share of variation in migration rates.

These findings suggest that the structural determinants of migration in India are largely persistent, shaped by long-run differences across states rather than by shorter-run changes within states over time. The results are interpreted as associations rather than causal effects..

I. Introduction

Migration is a universal phenomenon. It shapes the socio-economic and political environment at both national and regional levels and is an essential component of population change. Unlike fertility and mortality, migration is not a biological process. It is shaped by economic, social, and cultural forces, and influences the size, composition, and distribution of populations across space.

India presents a particularly compelling context for studying migration. Persistent disparities in income, employment opportunities, and urban development have produced large and sustained flows of inter-state migrants from relatively poor states like Bihar and Uttar Pradesh toward economically dynamic destinations like Maharashtra, Gujarat, and Karnataka. Understanding what drives these flows matters not just academically, but for how policymakers think about regional development and labour market policy.

This paper examines the determinants of inter-state migration in India using state-level panel data spanning three Census years 1991, 2001, and 2011. The central question is simple but underexplored: do the factors associated with migration differ depending on whether we look across states or within states over time? Two complementary estimators are employed to answer this. A between-state estimator identifies associations from persistent cross-state differences in income, urbanization, and density. A two-way fixed effects estimator identifies associations from within-state changes in labour market conditions, urbanization, and public expenditure over time. The objective is not to establish causal relationships but to document whether these two dimensions of variation tell the same story or different ones.

The paper addresses the following research question: Do the determinants of inter-state migration in India differ between states and within states over time?

Based on this, the following hypotheses are examined:

Between-state:

- H1: Higher per capita income is positively associated with migration across states.
- H2: Higher urbanization is positively associated with migration across states, though its independent effect may be limited once income is controlled for.
- H3: Population density has an ambiguous association with migration across states, reflecting opposing forces of agglomeration and congestion.

Within-state:

- H4: Increases in labour force participation within states are positively associated with migration over time.
- H5: Increases in urbanization within states are positively associated with migration over time.
- H6: Social sector expenditure and population density do not exhibit robust independent associations with migration in the within-state specification.

II. Literature Review

Migration in India has been widely studied in relation to regional disparities, labour markets, and structural transformation. Bhagat (2009) finds that inter-state migration increased relative to intra-state migration during the 1990s and is strongly associated with per capita income, sectoral composition, and rural poverty providing direct empirical motivation for the between-state analysis in the present study.

From a theoretical perspective, the Harris–Todaro model provides the foundational framework, emphasising expected income differentials rather than current labour market conditions as the key driver of migration decisions. This motivates H1 directly and explains why unemployment tends to perform poorly in empirical specifications relative to income-based measures.

Persistent cross-state differences in migration rates, however, cannot be fully explained by income alone. Munshi and Rosenzweig (2016) show that caste-based networks create significant mobility barriers, while Kone et al. (2018) document that language distance, cultural barriers, and state-specific entitlements further impede internal migration. Together, these findings suggest that persistent network and institutional differences across states absorbed by state fixed effects in the within estimator may account for substantial unexplained variation in migration rates.

On public expenditure, Imbert and Papp (2015) find that social spending primarily retains workers at the origin rather than attracting migrants to the destination consistent with the null within-state result on social expenditure documented here. The negative coefficient on social expenditure in the saturated between-state model may reflect that high-expenditure states tend to have slower structural transformation, though this should be interpreted cautiously given near-saturation concerns discussed in Section IV.

Building on this literature, the present study employs both between-state and within-state estimators to separately identify cross-sectional and time-series associations examining whether the determinants of migration operate differently at these two levels of variation.

III. Data and Methodology

A. Data Sources

Migration data in India have traditionally been collected by the Census based on place of birth. Since 1971, data have also been collected using place of last residence and duration of residence at the place of enumeration, allowing for a more accurate measure of recent migration flows. Internal migration can be classified into intra-district, inter-district, and inter-state migration, as well as by stream : rural-rural, rural-urban, urban-rural, and urban-urban.

The present study uses secondary data from three rounds of the Census of India - 1991, 2001, and 2011. Migration is measured using place of last residence data, focusing on individuals whose previous residence was in a different state from their current place of enumeration. Labour force participation is also constructed from Census data on main workers, marginal workers, and non-workers.

Additional data on population, urban population, population density, per capita NSDP, and social sector expenditure are obtained from the Reserve Bank of India's Handbook of Statistics on Indian States. These sources are combined to construct a state-level panel dataset covering three Census periods.

B. Sample Construction

The analysis is conducted at the state level, yielding a panel of 28 states across three time periods. Inter-state migration is the focus throughout. Union Territories namely Andaman and Nicobar Islands, Chandigarh, Daman and Diu, Dadra and Nagar Haveli, Delhi, Puducherry, and Lakshadweep are excluded due to missing data and structural differences in administrative and economic characteristics.

Observations with missing values in key variables are dropped at the state–year level. Three states namely Chhattisgarh, Jharkhand, and Uttarakhand were carved out of existing states in November 2000 and therefore do not exist as separate administrative units in the 1991 Census. All three enter the panel from 2001 onward where data are available. States with fewer than two usable observations are excluded to ensure meaningful within-state variation for panel estimation. The final dataset is an unbalanced panel comprising 28 states and 77 state-year observations, with between one and three observations per state across the three Census periods.

C. Variable Construction

The dependent variable is the inter-state migration rate, defined as the number of inter-state migrants divided by total state population. Population figures reported in thousands are converted to absolute values before constructing the rate. The migration rate is log-transformed in estimation to reduce skewness and improve interpretability of coefficients.

The explanatory variables are constructed as follows:

Labour force participation rate (LFPR) is defined as the sum of main workers and marginal workers divided by total population. This captures the share of the population engaged in economic activity and serves as a proxy for labour market conditions at the destination, replacing unemployment rates which are available only for non-Census years and therefore cannot be matched cleanly to the panel.

Urbanization rate is the share of urban population in total population, sourced directly from Census data for each period.

Per capita NSDP is constructed from state-level Net State Domestic Product at constant prices, available in three different base years 1980-81, 1999-00, and 2011-12 corresponding to the three Census periods. To construct a comparable real income series, these are rebased to constant 2011-12 prices using national-level implicit price deflators derived from per capita Net National Income data reported in Table 1.1 of the Economic Survey of India. Specifically, the conversion factors are 9.99 for the 1980-81 base series and 2.14 for the 1999-00 base series, constructed as the ratio of per capita NNI at constant 2011-12 prices to per capita NNI at current prices in the respective base years. This approach assumes that relative state-level price movements are proportional to national price movements, a standard approximation in the absence of state-level deflators. Per capita NSDP is then log-transformed in estimation.

Social expenditure share is constructed as social sector expenditure divided by NSDP. Social expenditure is reported in billions and NSDP in lakhs; both are converted to a common unit before taking the ratio. Since social expenditure is recorded in nominal terms while NSDP is at constant prices, the share is not strictly comparable across Census years. However, as this distortion affects all states uniformly within each period, cross-state comparisons remain valid. Year fixed effects further absorb any aggregate time-varying bias introduced by the changing price base of the NSDP denominator.

Population density is population per square kilometre, log-transformed to account for skewness in its distribution across states.

D. Empirical Strategy

The empirical analysis employs two complementary estimators that decompose the total variation in migration rates into cross-sectional and time-series components.

The **between-state estimator** regresses state-level time means of the dependent variable on time means of the regressors. This identifies associations from persistent cross-state differences why some states consistently attract more migrants than others and is estimated sequentially to examine the stability of the income coefficient as additional controls are introduced.

The between-state model takes the form:

$$\bar{y}_i = \beta_1 \overline{log(NSDPpc)}_i + \beta_2 \bar{X}_i + \bar{\varepsilon}_i$$

where $\bar{y}_i$ is the time-mean log migration rate for state i and $\bar{X}_i$ is the vector of time-mean controls.

The **two-way fixed effects estimator** controls for time-invariant state characteristics and common time trends, identifying associations from within-state changes over time. The model takes the form:

$$log(MigrationRate_{st}) = \beta_1 LFPR_{st} + \beta_2 Urbanization_{st} + \beta_3 SocialExpShare_{st} + \beta_4 log(Density_{st}) + \alpha_s + \gamma_t + \varepsilon_{st}$$

where α_s represents state fixed effects and γ_t represents time fixed effects. Standard errors are clustered at the state level to account for within-state serial correlation.

Per capita NSDP is included in the between-state specification but excluded from the within-state specification. State fixed effects absorb all time-invariant state characteristics including persistent income levels leaving only within-state income growth to identify the coefficient. Given that income levels are highly persistent and within-state growth was broadly similar across states over this period, income adds limited independent variation in the within estimator and is more cleanly examined in the between-state framework.

Both specifications are interpreted as capturing associations rather than causal effects. The between estimator is subject to omitted variable bias from unobserved cross-state differences, while the within estimator addresses time-invariant confounders but cannot rule out time-varying omitted variables or reverse causality between migration and the regressors.

IV. Empirical Analysis

A. Model Specification

The empirical analysis employs two complementary estimators: a between-state estimator and a two-way fixed effects estimator as described in the methodology section. The dependent variable throughout is the logarithm of the inter-state migration rate, defined as the number of inter-state migrants relative to total state population.

The between-state estimator is estimated sequentially, adding regressors progressively to examine the stability of the income coefficient across specifications. The within-state fixed effects model is estimated as a single specification:

$$log(MigrationRate_{st}) = \beta_1 LFPR_{st} + \beta_2 Urbanization_{st} + \beta_3 SocialExpShare_{st} + \beta_4 log(Density_{st}) + \alpha_s + \gamma_t + \varepsilon_{st}$$

The panel is unbalanced. Chhattisgarh, Jharkhand, and Uttarakhand were created in November 2000 and therefore do not exist as separate administrative units in the 1991 Census. All three enter the panel from 2001 onward where data are available. The final sample comprises 28 states and 77 state-year observations in the within-state specification, and 28 state means in the between-state specification.

B. Multicollinearity Diagnostics

Multicollinearity is assessed using pairwise correlation analysis and Variance Inflation Factor tests. The correlation matrix reveals moderate negative correlation between labour force participation and log population density (-0.582), and moderate positive correlation between log per capita NSDP and urbanization (0.577). The latter is particularly relevant for interpreting the between-state results, as it indicates that income and urbanization capture overlapping dimensions of economic development across states. A notable negative correlation between social expenditure share and log per capita NSDP (-0.504) reflects that poorer states tend to allocate a higher share of output to social spending.

VIF values for all explanatory variables remain below 3 in both specifications, falling well within conventional thresholds. Multicollinearity is therefore not a mechanical concern, though the correlation between income and urbanization has substantive implications for interpreting the between-state sequential results, as discussed below.

C. Between-State Results

The between-state estimator identifies associations from persistent cross-state differences in migration rates. Models are estimated sequentially to examine whether the income coefficient is stable as additional controls are introduced.

The results are presented in Table A2(Appendix). In the income-only specification, per capita NSDP is positive and highly significant ($\beta = 1.468$, $p < 0.001$), explaining approximately 53.0 percent of cross-state variation in migration rates. This is a strong and clean result stating richer states attract substantially more migrants, consistent with H1 and with the Harris–Todaro framework.
Adding urbanization and density sequentially leaves the income coefficient virtually unchanged, ranging narrowly between 1.468 and 1.834 across Models 1 through 3. Neither urbanization nor density achieves statistical significance when entered alongside income, and the gain in between R^2 from adding both variables is modest rising from 0.530 to 0.599. Income is therefore not merely proxying for urbanization or density; it operates as an independent predictor of cross-state migration patterns. H2 and H3 are not supported in the parsimonious specifications.
When LFPR and social expenditure are added individually alongside income, urbanization, and density (Models 3b and 3c), the income coefficient remains stable and significant in both cases 1.779 and 1.661 respectively. Neither LFPR nor social expenditure achieves statistical significance when added individually.
The fully saturated model which includes all five regressors simultaneously produces a markedly different picture. The income coefficient falls to 0.312 and becomes insignificant. Social expenditure enters negatively and significantly ($\beta = -4.795$, $p < 0.05$), and log density becomes significantly negative ($\beta = -0.708$, $p < 0.01$).
The between R^2 jumps sharply to 0.964. This pattern i.e. a large R^2 increase alongside coefficient instability is characteristic of near-saturation with N=28 rather than a genuine change in the underlying relationships. With five regressors and 28 observations, the design matrix is insufficiently supported for stable coefficient estimation, and the fully saturated model should not be interpreted at face value.
The negative social expenditure coefficient in the saturated model is directionally consistent with Imbert and Papp (2015) states with higher social expenditure shares may be those with slower structural transformation and less dynamic labour markets but cannot be cleanly identified in this framework.
The parsimonious and reliable between-state finding is that income is the dominant cross-state predictor of migration, consistent with H1. The finding is robust across sequential specifications through Model 3c and does not depend on controlling for urbanization, density, LFPR, or social expenditure individually.

D. Within-State Results

The two-way fixed effects estimator identifies associations from within-state changes over time, controlling for all time-invariant state characteristics and common time trends. Results are presented in Table A1.

Labour force participation exhibits a positive and statistically significant coefficient ($\beta = 3.051$, $p < 0.01$). A 10 percentage point increase in LFPR within a state is associated with approximately 30.5 percent higher migration rates, suggesting that expanding labour markets attract migrants over time. This is consistent with H4.

Urbanization is also positive and statistically significant ($\beta = 2.000$, $p < 0.01$), indicating that increases in urban population share within states are associated with higher migration rates. A 10 percentage point increase in urbanization is associated with approximately 20 percent higher migration rates, consistent with H5 and with the role of urban expansion as a pull factor for migrants.

Social expenditure share and log population density are both statistically insignificant in the within-state specification, with p-values of 0.374 and 0.314 respectively. The social expenditure coefficient is positive but imprecisely estimated, consistent with H6 and with the interpretation that destination-side public spending does not independently attract migrants. This is in line with Imbert and Papp (2015), who find that social spending primarily affects migration at the origin rather than the destination.

The pooled R^2 is 0.284, but the within-group R^2 is -0.082 meaning the regressors explain less within-state variation than state and time fixed effects alone. This indicates that state fixed effects absorbing persistent cross-state differences in income, geography, institutions, and migration networks account for the dominant share of variation in migration rates. Time-varying changes within states explain comparatively little additional variation once this heterogeneity is controlled for. This is itself an informative finding, consistent with Munshi and Rosenzweig (2016), who document that persistent network and institutional structures mediate migration decisions in ways that attenuate the role of time-varying economic signals.

Per capita NSDP is not included in the within-state specification. State fixed effects absorb time-invariant income differences across states, and within-state income growth was broadly similar across states over this period, contributing limited independent identifying variation. Its role is examined in the between-state framework.

E. Robustness

A lagged specification is estimated to address potential reverse causality, the concern that higher migration itself raises labour force participation or urbanization at the destination. Due to the limited time dimension of the panel (T=3), lagging the regressors reduces the sample to 49 observations and leaves only one cross-sectional difference per state for identification. The lagged model yields uniformly insignificant results within R^2 of -0.512, reflecting insufficient statistical power rather than a genuine absence of association. This specification is therefore not emphasised but is reported in Appendix A7 for completeness.

The between-state results are robust to the sequential addition of controls through Models 1 to 3c, with the income coefficient remaining stable in sign, magnitude, and significance throughout, ranging from 1.468 to 1.839. This stability across six specifications provides confidence that the between-state income finding is not an artefact of omitted variable bias from the included controls.

V. Conclusion

The present study examines the determinants of inter-state migration in India using state-level panel data across three Census years, employing both between-state and within-state estimators to separately identify cross-sectional and time-series associations.

Between states, per capita income emerges as the dominant and robust predictor of migration. The income coefficient remains positive, large, and statistically significant across six sequential specifications, explaining approximately 53 percent of cross-state variation in migration rates when entered alone. Neither urbanization, population density, labour force participation, nor social expenditure adds meaningful independent explanatory power when entered alongside income in parsimonious specifications. This finding is consistent with the Harris–Todaro framework and with Bhagat (2009), and confirms that persistent income differentials across Indian states are the primary structural driver of who attracts migrants and who does not.

Within states over time, the picture is different. After controlling for time-invariant state characteristics through fixed effects which absorb persistent income, network, and institutional differences, labour force participation and urbanization emerge as the significant within-state correlates of migration. A 10 percentage point increase in LFPR or urbanization within a state is associated with approximately 30 and 20 percent higher migration rates respectively. Social expenditure and population density do not exhibit robust within-state associations, consistent with the view that destination-side fiscal spending is not a meaningful pull factor for migrants.

The contrast between the two sets of results is itself the central finding. The near-zero within-group R^2 indicates that time-varying changes within states explain very little migration variation once persistent state-level heterogeneity is accounted for. The structural determinants of migration in India appear to be largely persistent and cross-sectional, established by long-run income and development differences across states rather than driven by shorter-run within-state changes in labour market conditions or public expenditure.

The study has several limitations. Neither estimator establishes causal relationships; the between estimator is subject to omitted cross-state confounders, and the within estimator cannot rule out time-varying omitted variables or reverse causality. The absence of district-level data prevents examination of within-state heterogeneity, and the three-period panel limits the statistical power of the within estimator. Additionally, factors such as migration networks, language barriers, and institutional constraints documented by Munshi and Rosenzweig (2016) and Kone et al. (2018) are not explicitly modelled and may account for a substantial share of the unexplained variation. Future work extending this analysis to district-level data, longer panels, or quasi-experimental identification strategies would substantially strengthen the causal interpretation of these findings.

I. References

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Appendix

A1. Full Regression Results (Two-Way Fixed Effects)

Dependent variable: Log(Migration Rate)

Variable	Coefficient	Std. Error	t-stat	p-value
LFPR	3.051	0.985	3.098	0.004
Urbanization	2.000	0.662	3.020	0.004
Social Exp Share	0.335	0.373	0.898	0.374
Log Density	0.543	0.533	1.019	0.314
Log Per Capita NSDP	0.091	0.139	0.658	0.514
Constant	-9.085	3.688	-2.463	0.018

Model statistics:

- Observations: 77
- Number of states: 28
- Time periods: 3 (1991, 2001, 2011)
- R^2 (Pooled): 0.284
- R^2 (Within): -0.082
- R^2 (Between): -0.469
- R^2 (Overall): -0.533
- Fixed effects: State and Year
- Standard errors: Clustered at state level

A2. Between-State Sequential Results

Dependent variable: Log(Migration Rate) - state time means

	BE(1)	BE(2)	BE(3)	BE(3b)	BE(3c)	BE(4)
Log NSDPpc	1.468***	1.834***	1.777***	1.779***	1.661***	0.312
	(0.271)	(0.321)	(0.335)	(0.341)	(0.347)	(0.211)
Urbanization	-	-2.204†	-1.996	-1.953	-2.008†	0.802
		(1.149)	(1.200)	(1.244)	(1.191)	(1.393)
Log Density	-	-	-0.077	-0.097	-0.187	-0.708***
			(0.112)	(0.153)	(0.146)	(0.183)
LFPR	-	-	-	-0.575	-	-5.311
				(2.942)		(3.584)
Social Exp Share	-	-	-	-	-1.812	-4.795**
					(1.563)	(1.899)
Constant	-18.817***	-22.090***	-21.134***	-20.819***	-18.999***	-
Between R^2	0.530	0.591	0.599	0.599	0.621	0.964
N (states)	28	28	28	28	28	28

Standard errors in parentheses. †$p<0.10$, *$p<0.05$, **$p<0.05$, ***$p<0.01$

Note: BE(4) includes all five regressors simultaneously. The large R^2 increase from BE(3c) to BE(4) alongside sign reversals and coefficient instability, particularly the urbanization sign flip and the large negative LFPR coefficient, are consistent with near-saturation given N=28. The parsimonious and reliable between-state finding is from BE(1). The negative and significant social expenditure coefficient in BE(4) is directionally consistent with Imbert and Papp (2015) but should be interpreted cautiously given the saturation concern.

A3. Variable Definitions

Variable	Definition	Source	Transformation
Migration Rate	Inter-state migrants / Total population	Census of India	Log
LFPR	(Main workers + Marginal workers) / Population	Census of India	Proportion
Urbanization	Urban population / Total population	Census of India	None
Log Per Capita NSDP	NSDP at constant 2011-12 prices / Population	RBI Handbook	Log
Social Exp Share	Social sector expenditure / NSDP	RBI Handbook	Ratio
Log Density	Population per sq. km	RBI Handbook	Log

A4. Data Construction and Cleaning

- Union Territories excluded: Andaman and Nicobar Islands, Chandigarh, Daman and Diu, Dadra and Nagar Haveli, Delhi, Puducherry, and Lakshadweep
- States retained based on having at least two usable observations across the three Census years
- Chhattisgarh, Jharkhand, and Uttarakhand were created in November 2000 and do not have 1991 data as separate administrative units; all three are retained for 2001 and 2011 where data are available
- Population converted from thousands to absolute values before constructing migration rate
- Social expenditure reported in billions and NSDP in lakhs; both converted to a common unit (multiplied by 10^9 and 10^5 respectively) before taking the ratio
- Per capita NSDP rebased to constant 2011-12 prices using national implicit price deflators: conversion factor 9.99 for the 1980-81 base series and 2.14 for the 1999-00 base series
- Observations with missing values in any key variable dropped at the state-year level
- Final dataset: unbalanced panel, 77 state-year observations, 28 states

A5. Correlation Matrix

	LFPR	Urbanization	Social Exp	Log Density	Log NSDPpc
LFPR	1.000	0.115	0.035	-0.582	0.290
Urbanization	0.115	1.000	-0.327	0.202	0.577
Social Exp Share	0.035	-0.327	1.000	-0.462	-0.504
Log Density	-0.582	0.202	-0.462	1.000	0.014
Log NSDPpc	0.290	0.577	-0.504	0.014	1.000

A6. Variance Inflation Factors

Variable	VIF
LFPR	1.817
Urbanization	1.624
Social Exp Share	1.923
Log Density	2.328
Log Per Capita NSDP	2.122

Note: All VIF values are below conventional thresholds, indicating no significant multicollinearity.

A7. Lagged Specification -Robustness Check

Dependent variable: Log(Migration Rate)

Variable	Coefficient	Std. Error	p-value
LFPR (lag)	0.525	3.046	0.865
Urbanization (lag)	2.176	3.403	0.532
Social Exp Share (lag)	0.523	0.712	0.473
Log Density (lag)	0.619	1.089	0.578
Log Per Capita NSDP (lag)	0.680	0.584	0.261
Constant	-14.464	10.947	0.205

Model statistics:

- Observations: 49
- Number of states: 27
- Time periods: 2
- R^2 (Within): -0.512
- Fixed effects: State and Year
- Standard errors: Clustered at state level

A8. F-test for Poolability

Specification	F-statistic	p-value	Conclusion
Two-Way FE (main)	38.621	0.000	State fixed effects jointly significant
Lagged specification	20.431	0.000	State fixed effects jointly significant

Note: The F-test for poolability strongly rejects the null hypothesis that state fixed effects are jointly zero in both specifications, confirming that the fixed effects estimator is preferred over pooled OLS.